Vaijayanth Sheri *M. Eng*

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LinkedIn ↗ | Portfolio ↗ | GitHub ↗

PROFILE

- Product-minded Energy Systems Engineer focused on bridging the gap between complex energy modeling (PyPSA, OEMOF) and scalable B2B software solutions.
- Proven experience building automated, user-friendly tools and dashboards that accelerate spatial dimensioning, financial feasibility analysis, and energy system planning.
- Deep technical understanding of the German commercial energy market, including PV/battery optimization, load profile analysis, and asset simulation.
- Strong Python developer dedicated to owning the product lifecycle from physical energy equations to front-end sales enablement interfaces.

EDUCATION

Masters in Hydrogen Technology, *Technischen Hochschule Rosenheim* — 10.2025 – Present | Burghausen

Masters in Renewable Energy Systems, *Hochschule Nordhausen* — 10.2020 – 03.2024 | Nordhausen, Germany

STUDENT EXPERIENCE

Working Student - Utility Infrastructure Asset Simulation, — 11.2025 – Present | München
Stadtwerke München

- Processed and validated large-scale electricity network datasets (assets, operational profiles) using PostgreSQL to build a reliable backbone for infrastructure planning.
- Prepared and structured complex time-series data for asset lifecycle and cost simulation models, directly supporting 1–50 year planning horizons.
- Identified and corrected inconsistencies in legacy database systems to ensure the downstream usability of data for capacity expansion and operational logic.

Integration of Open-Source Tools in GUI for Energy System Modelling, — 08.2023 – 03.2024 | Nordhausen, Germany
Master Thesis at Hochschule Nordhausen

- Developed custom PyQt5 graphical interfaces for PVLib, WindPowerLib, and DemandLib, enabling non-technical users to independently run complex energy profile simulations.
- Integrated generated data directly into OEMOF-based energy system models to validate optimization scenarios, reducing data preparation time by 20%.
- Translated complex physical modeling workflows into intuitive, zero-code software tools, mirroring the need for scalable sales enablement in energy tech.

Intern as DATA Scientist, *Techno-Co-labs Software Inc.* — 05.2023 – 07.2023 | India

- Built end-to-end data pipelines and exploratory analytics workflows in Python, processing large-scale business datasets to accelerate insight delivery.
- Designed predictive machine learning models to enhance business metric forecasting, improving classification accuracy by 15% through robust feature engineering.
- Delivered data-driven visualizations and reports to cross-functional stakeholders, translating technical outputs into actionable business intelligence.

PROJECTS

PyPSA Results Visualization Dashboard, *Python, PyPSA, Plotly, xarray* — 09.2025

- Developed a no-code web application to parse, aggregate, and instantly visualize multi-dimensional PyPSA grid simulation outputs, eliminating manual script execution.
- Engineered interactive geographic network maps and time-series plots to evaluate differing network topologies, dispatch outcomes, and KPIs side-by-side.
- Optimized data pipelines using Pandas and xarray to handle large netCDF datasets dynamically, significantly accelerating exploratory scenario analysis.

Solar + EV Configurator, *Python, Streamlit, PVGIS API, Shapely* 07.2025

- Architected a full-stack spatial and financial modeling application enabling rapid dimensioning of PV and EV assets for scalable EaaS customer acquisition.
- Engineered an interactive geospatial pipeline linked with the PVGIS API to calculate exact roof capacity and localized dynamic solar yields.
- Built a financial engine to evaluate battery sizing, self-consumption, and feed-in tariffs, programmatically generating PDF payback reports for decision-making.

Hybrid PV+Wind+Storage Digital Twin, 06.2025
Python, Pandas, OpenWeatherMap API

- Built an interactive simulation sandbox that balances real-time weather forecasts and standard BDEW H0 load profiles against hybrid renewable generation models.
- Implemented vectorized physical power equations and chronological battery state-of-charge logic to predict real-time grid reliance and microgrid autonomy.
- Designed a responsive Streamlit dashboard featuring multi-axis time-series visualization for rapid assessment of energy flow and peak load behavior.

TECHNICAL SKILLS

- **Energy System Modeling** — PyPSA, OEMOF, Battery Storage Modeling, PVLib, WindPowerLib
- **Python Ecosystem** — Python, Pandas, NumPy, xarray, scikit-learn
- **Software Engineering** — Streamlit, PyQt5, API Integration, Git, Full Stack Development
- **Grid & Spatial Analytics** — Load Profile Analysis (Lastgänge), Peak Shaving (Lastspitzenkappung), GIS Mapping (Folium), Time-Series Analysis, BDEW Standard Profiles
- **Product & Business** — B2B SaaS Product Management, Technical Sales Enablement, Energy-as-a-Service (EaaS), Financial ROI Modeling, Cross-functional Communication

COURSES & CERTIFICATIONS

Time Series Analysis: Forecasting Models with Python, *Udemy* 07.2025

Learning Data Analytics: 1 Foundations, *LinkedIn Learning* 03.2025

Power BI full course, *Learnit Training* 05.2024

Machine Learning with Python, *Cognitive class* 05.2023

Python for Data Science, *Cognitive class* 04.2023

Agile project management, *OPENCLASSROOMS* 12.2022

LANGUAGES

- **English** (Business fluent - C1)
- **German** (Good - B1)
- **Telugu** (Native)
- **Hindi** (Bilingual)

Vaijayanth Sheri
München, 31.05.2026

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31 May 2026

Enerkii GmbH
Frauenlobstraße 2
80337 München

Subject: Application for Energy Product Manager

Dear Ms Osterhammer,

Enerkii's mission to scale the Energy-as-a-Service model for German mid-sized businesses relies heavily on the capabilities of enerkii OS. The core challenge is bridging the gap between complex energy system modeling and the rapid, user-friendly insights required by your sales teams. My background in building Python-based energy simulation GUIs and data pipelines positions me to take ownership of enerkii OS and translate deep technical domain knowledge into scalable sales enablement features.

I am an energy systems modeler and software developer with practical experience in both utility data infrastructure (Stadtwerke München) and open-source tool development (PyPSA, OEMOF). My work consistently focuses on abstracting complex mathematical models—such as solar yield forecasting and system optimization—into intuitive, productized dashboards that empower non-technical users to make data-driven decisions.

Here is how my experience aligns with the strategic goals for enerkii OS:

Energy System Modeling & PyPSA Architecture

- **Evidence:** Developed a Streamlit-based PyPSA Code Generator with real-time semantic validation and a PyPSA Results Visualization Dashboard to automate post-run analysis of multi-dimensional .nc datasets.
- **Relevance:** I can directly accelerate the technical development of enerkii OS, integrating complex PyPSA constraints while ensuring robust data validation for accurate load profile analysis and peak shaving simulations.

Translating Complex Models for Sales Enablement

- **Evidence:** Engineered a full-stack Solar + EV Configurator that combines user-drawn rooftop polygons, PVGIS data, and dynamic financial engines (payback periods, self-consumption) to instantly generate automated PDF reports.
- **Relevance:** I understand how to abstract complex parameters into a clean UX, allowing your AEs and SDRs to rapidly generate accurate, localized savings projections and close B2B deals without bottlenecking the engineering team.

Data Integrity for Infrastructure Planning

- **Evidence:** At Stadtwerke München, I managed and validated large, multi-source electricity network datasets using PostgreSQL, resolving time-series anomalies for asset simulation models.
- **Relevance:** Reliable projections require clean data. I can ensure that the real-world operational data feeding into enerkii OS remains consistent, supporting reliable arbitrage and strategic purchasing decisions.

Commercial Pragmatism & Product Ownership

- **Evidence:** Designed tools driven by comprehensive Product Requirements Documents (PRDs) and conducted feasibility modeling for co-located PV and battery assets to support financial business cases.
- **Relevance:** I do not just write code; I prioritize features based on commercial impact. I can effectively manage the product roadmap by balancing the technical debt of energy models with the immediate feature requests of the sales floor.

I am eager to bring my product mindset and Python modeling expertise to enerkii, helping to scale your EaaS customer acquisition engine. I look forward to discussing how my dual expertise in energy engineering and software development can accelerate the roadmap for enerkii OS.

Sincerely,
Vaijayanth Sheri